

# LeetCode 815: Bus Routes

Optimal BFS Solution on Multi-Graph / Stop-to-Bus Mapping

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Language: C++

## Key Approach & Intuition

### 1. Mapping Creation

Map each bus stop to all the bus routes that pass through it:

```
unordered_map<int, vector<int>> mp
```

### 2. Level-order BFS

Perform BFS starting from `source` stop. Track visited buses and visited stops using hash sets to prevent redundant processing.

## C++ Source Code Implementation

```
class Solution {
public:
    int numBusesToDestination(vector<vector<int>>& routes, int source, int target) {
        int n = routes.size(); //number of busses;

        //kis stop se kaun kaun si bus le skte
        unordered_map<int, vector<int>> mp;

        for(int i = 0; i < n; i++){
            vector<int> stops = routes[i];
            for(int j = 0; j < stops.size(); j++){
                mp[stops[j]].push_back(i);
            }
        }

        queue<int> q; //bas current stop rkhenge aur har bus me jaayenge
        unordered_set<int> seenbus;
        unordered_set<int> seenstop;

        q.push({source});
        seenstop.insert(source);
        int bus = 0;
        while(!q.empty()){
            int currsz = q.size();

            while(currsz--){
                int currstop = q.front();
                q.pop();
                if(currstop == target) return bus;

                for(auto &bus : mp[currstop]) //saare bus iss stop ke
                {
```

```

        if(seenbus.find(bus) != seenbus.end()) continue;

        seenbus.insert(bus);
        for(auto &station : routes[bus]){ //iss bus se har stop pe jaayenge
            if(seenstop.find(station) == seenstop.end()){
                seenstop.insert(station);
                q.push({station});
            }
        }
    }
}

bus++; //iss station se har bus me baith gaye
}

return -1;
}
};

```

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